

Linear Algebra and Calculus Lecture 10

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Properties of Determinants

We are going to see some of the things that we can do with determinants.

Theorem. (*Row Operations*) Let A be a square matrix.

1. If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$.
2. If two rows of A are interchanged to produce B , then $\det B = -\det A$.
3. If one row of A is multiplied by k to produce B , then $\det B = k \det A$.

Exercise. Compute $\det A$, where $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}$

Theorem. A square matrix A is invertible if and only if $\det A \neq 0$.

Theorem. If A is an $n \times n$ matrix, then $\det A^T = \det A$.

Theorem. If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

Exercise. For an invertible matrix A , show that $\det(A^{-1}) = \frac{1}{\det A}$.

Vector Spaces

Until now we have only seen vectors in $\mathbb{R}^n$, but not all vectors are lists of numbers. Here we present a more generalized definition.

Definition. A **vector space** is a nonempty set V of objects, called *vectors*, on which are defined two operations, called *addition* and *multiplication by scalars*, subject to the ten axioms (or rules) listed below. The axioms must hold for all vectors u, v and w in V and for all scalars c and d .

1. The sum of u and v , $u + v$, is in V .
2. $u + v = v + u$
3. $(u + v) + w = u + (v + w)$
4. There is a zero vector 0 in V , and $u + 0 = u$
5. For each u in V , there is a vector $-u$ in V such that $u + (-u) = 0$
6. The scalar multiple of u by c , cu , is in V
7. $c(u + v) = cu + cv$
8. $(c + d)u = cu + du$
9. $c(du) = (cd)u$
10. $1u = u$

Example. For $n \geq 0$, the set $\mathbb{P}_n$ of polynomials of degree at most n consists of all polynomials of the form

$$p(t) = a_0 + a_1t + a_2t^2 + \cdots + a_nt^n,$$

where the coefficients $a_0, \dots, a_n$ and the variable t are real numbers. To prove $\mathbb{P}_n$ is a vector space it suffices to show the closure under addition and scalar multiplication. Indeed, we know from calculus that if p and q are polynomials, $p + q$ is also a polynomial. And if $c \in \mathbb{R}$, then cp is also a polynomial, since

$$(cp)(t) = cp(t) = ca_0 + (ca_1)t + \cdots + (ca_n)t^n.$$

Exercise. Show that the set V of all real-valued functions defined over $\mathbb{R}$ is a vector space.

Subspaces

Definition. A subspace of a vector space V is a subset H of V that has three properties:

1. The zero vector of V is in H .
2. H is closed under vector addition. That is, for each u and v in H , the sum $u + v$ is in H .
3. H is closed under multiplication by scalars. That is, for each u in H and each scalar c , the vector cu is in H .

Example. The space of polynomials $\mathbb{P}_n$ is a subspace of the space of all real-valued functions.

Exercise. Show that the following subset of $\mathbb{R}^3$ is a subspace:

$$H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} \in \mathbb{R}^3 : x_1, x_2 \in \mathbb{R} \right\}.$$

Subspace Spanned by a Set

Theorem. If $v_1, \dots, v_p$ are in a vector space V , then $\text{Span}\{v_1, \dots, v_p\}$ is a subspace of V .

Proof. Let's write the proof for $p = 2$ for clarity purposes, even though it works the same for any p . Given v_1 and v_2 in a vector space V , let $H = \text{Span}\{v_1, v_2\}$. The zero vector is in H , since $0 = 0v_1 + 0v_2$. To show that H is closed under vector addition, take two arbitrary vectors in H , say,

$$u = s_1v_1 + s_2v_2 \quad \text{and} \quad w = t_1v_1 + t_2v_2.$$

By Axioms 2, 3, and 8 for the vector space V ,

$$\begin{aligned}u + w &= (s_1v_1 + s_2v_2) + (t_1v_1 + t_2v_2) \\&= (s_1 + t_1)v_1 + (s_2 + t_2)v_2.\end{aligned}$$

So $u + w$ is in H . Furthermore, if c is any scalar, then by Axioms 7 and 9,

$$cu = c(s_1v_1 + s_2v_2) = (cs_1)v_1 + (cs_2)v_2$$

which shows that cu is in H and H is closed under scalar multiplication. Thus, H is a subspace of V . $\square$

Exercise. Let H be the set of all vectors of the form $(a - 3b, b - a, a, b)$, where a and b are arbitrary scalars. That is, let $H = \{(a - 3b, b - a, a, b) \mid a \text{ and } b \in \mathbb{R}\}$. Show that H is a subspace of $\mathbb{R}^4$.